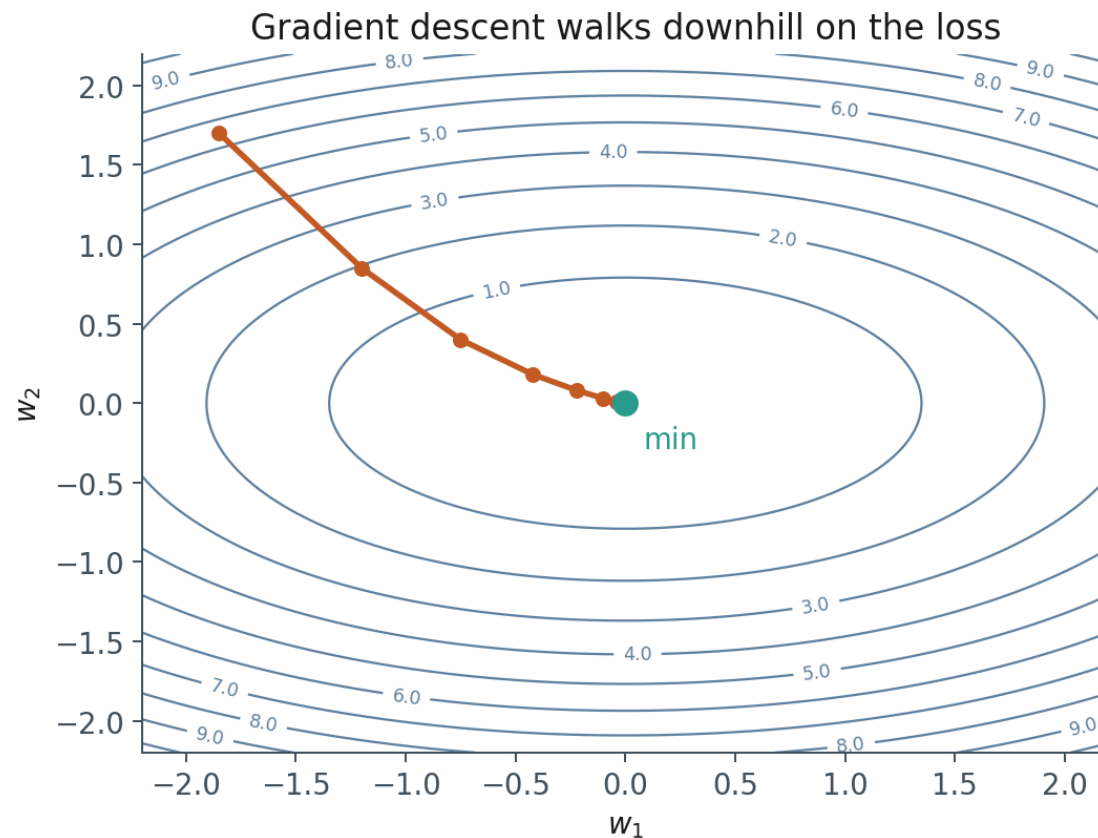


Learning goals for this block

- Write $\theta \leftarrow \theta - \eta \nabla L$ and take two numeric steps.
- Name the four boxes: forward, loss, backward, update.
- Derive $\partial L / \partial w = (a - y) \text{ReLU}'(z) x$ for one unit.
- Pass logits z into BCEWithLogitsLoss, not sigmoid(z).

Training is walking downhill

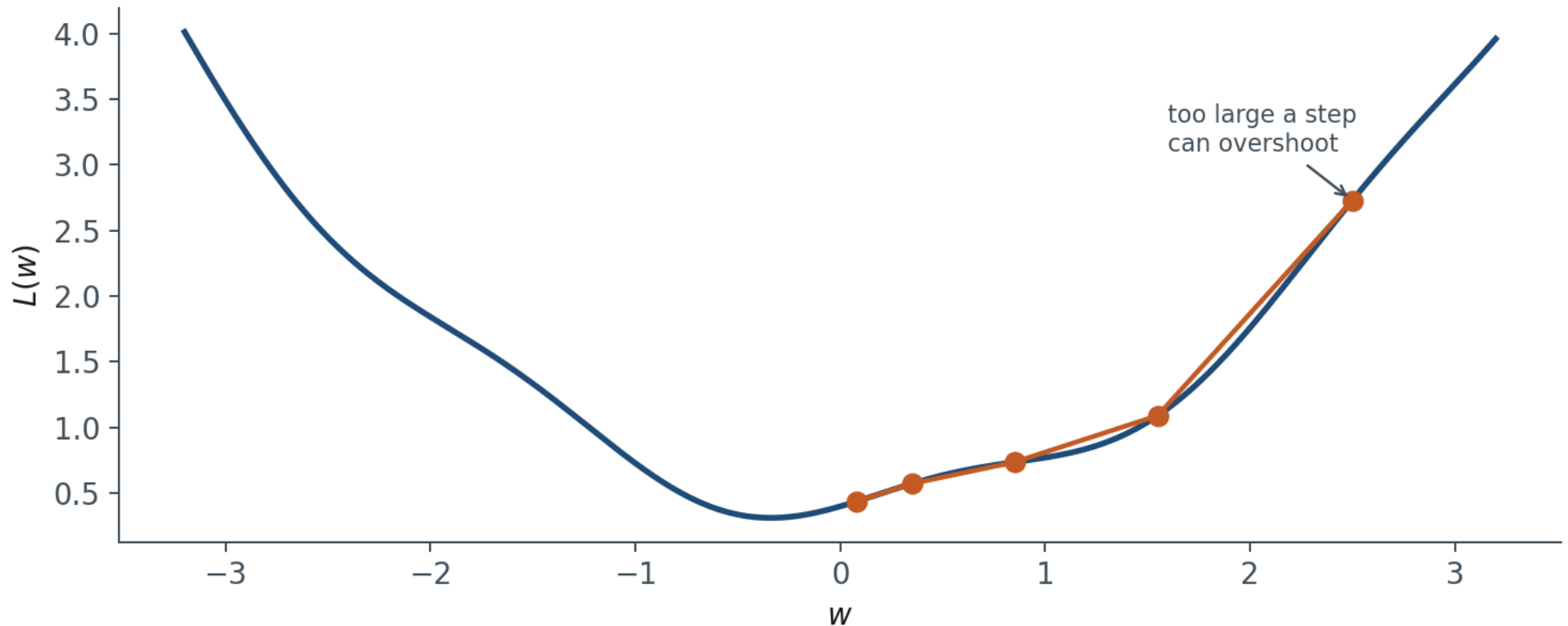
- Pack weights into θ . Loss $L(\theta)$ is a scalar.
- Classical names: loss = cost = empirical risk.
- The gradient points uphill. We step the other way.



The update and the learning rate

- $\theta \leftarrow \theta - \eta \nabla L(\theta)$.
- Too small: crawl. Too large: overshoot.
- SGD estimates the gradient on a minibatch. Adam waits.
- Nielsen Ch. 1: this picture, then stop before long MNIST.

Update: $w \leftarrow w - \eta \nabla L(w)$



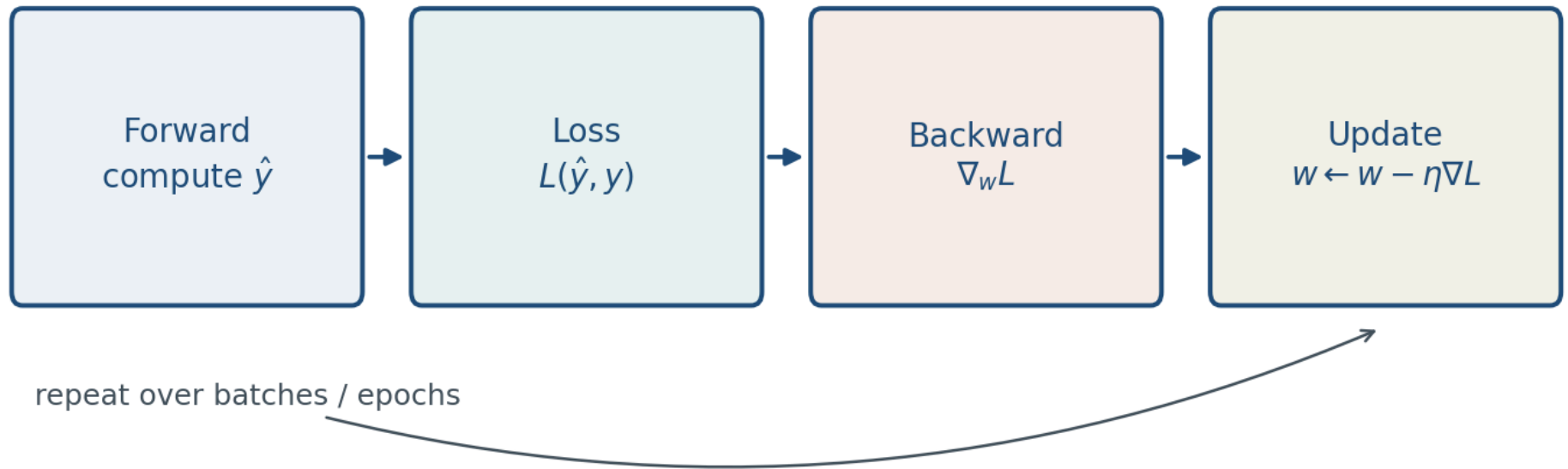
Two steps, then overshoot

- $L(\theta) = (\theta - 3)^2$, $\theta_0 = 0$, $\eta = 0.25$.
- $\theta_1 = 1.5$, $\theta_2 = 2.25$. Minimum is at 3.
- $\eta = 2$ sends $\theta_1 = 12$: overshoot, not a new algorithm.

Forward, loss, backward, update

- Keep activations: the backward pass needs them.
- Backprop is the chain rule. Autograd is backprop.
- Forget `zero_grad()`: gradients accumulate. Silent η disaster.
- Pretraining and fine-tuning are this loop with different data and L .

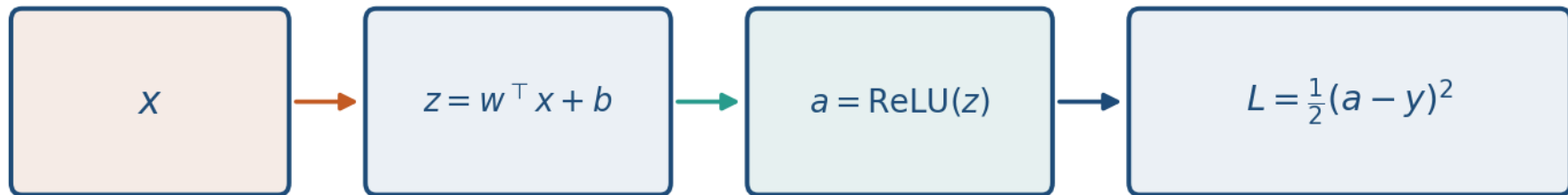
Training a net is this loop. Backprop is just the chain rule.



One ReLU unit as gates

- CS231N: each op is a gate. Local derivative times upstream.
- $z = w^\top x + b$, $a = \text{ReLU}(z)$, $L = \frac{1}{2}(a - y)^2$.
- If $z \leq 0$, this example does not move w .

One ReLU unit: cache x and z ; autograd is this chain



backward: $(a - y) \cdot 1_{z > 0} \cdot x$ (CS231N: local $\times$ upstream)

The chain rule you must write

- $\partial L / \partial w = (a - y) \text{ReLU}'(z) x$.
- CMU 11-711 cousin for softmax+CE: $\partial L / \partial w = (p - y)x$.
- Numeric: $x = 2, w = 0.5, y = 0$ gives $\partial L / \partial w = 2$; $\eta = 0.1$ sends $w \leftarrow 0.3$.

Lab 1 loss takes logits

- BCEWithLogitsLoss is binary cross-entropy on z , not on $\sigma(z)$.
- If you sigmoid first, you squash twice.
- Next-token NLL (note 1.1) is the same loop at vocabulary scale.
- 3Blue1Brown GD + backprop videos: homework, not this block.